Designing Breeding Programs

Breeding Programs Developing countries scenario

- Majority of the world's population lives in developing countries
- Above 75% of total population of developing world dwells in rural areas
- Agriculture in the form of smallholder farming and pastoralism forms the backbone of livelihood and economy
- Level of poverty is daunting
  - 2.1 billion people live on less than 2US$/day
  - 880 million people live on less than 1US$/day
- ~1 billion people (70% of the world's extremely poor) depend on livestock for livelihood

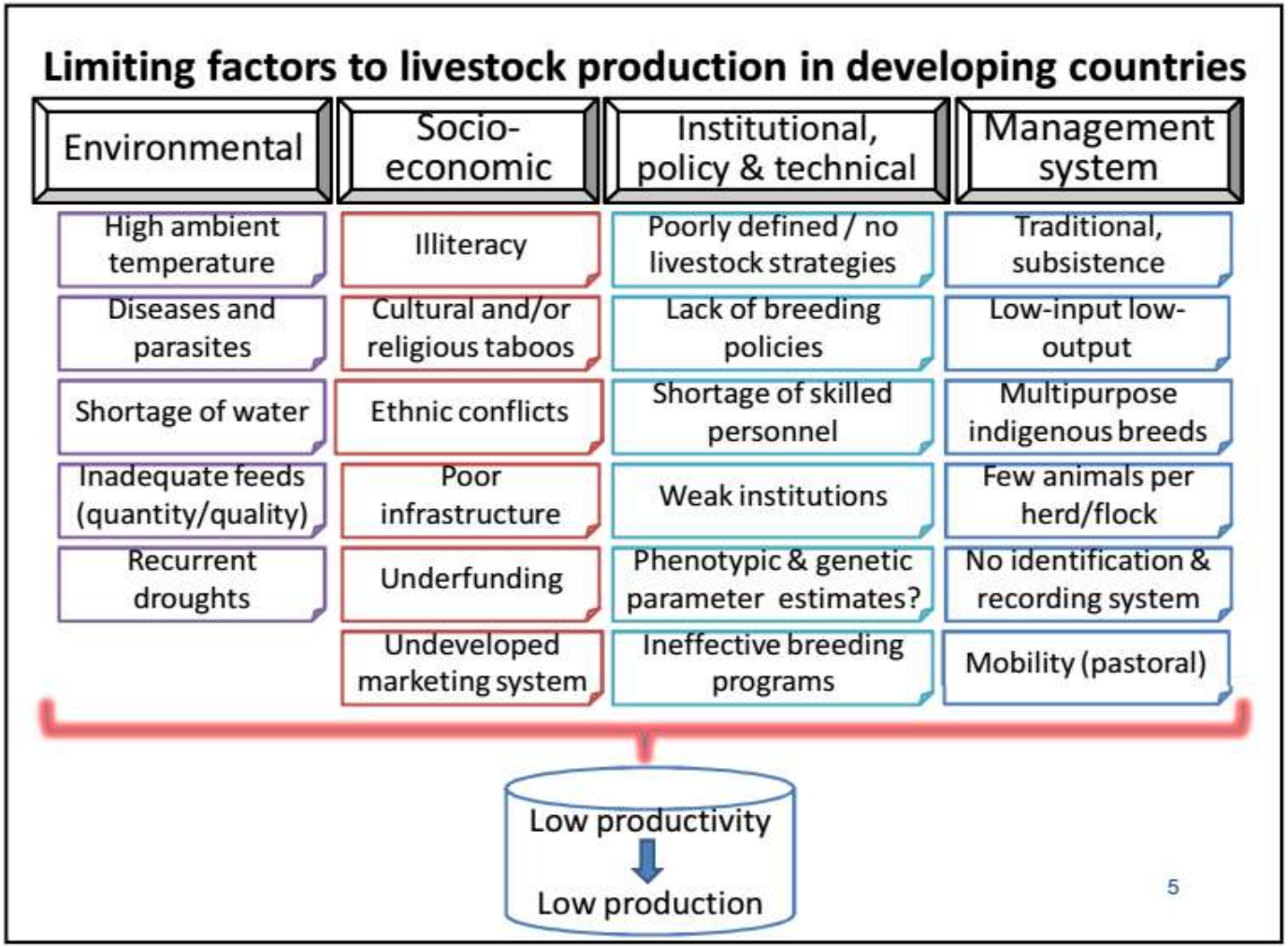
Breeding Programs Developing countries scenario

- Increased demands for animal products due to population growth, urbanization and rising income
  - National production levels are unable to satisfy the growing local demands
    - Sharp increases in import volumes (particularly dairy products)
- 77% of the global livestock breeds in active use are found in developing countries
- 94% of these breeds are not subjected to structured genetic improvement programs
  - Potential of livestock resources remained untapped

Per capita consumption of livestock products

Region	Meat (kg per year)			Milk (kg per year)		
	1964 - 1966	1997 - 1999	2030	1964 - 1966	1997 - 1999	2030
World	24.2	36.4	45.3	73.9	78.1	89.5
Developing countries	10.2	25.5	36.7	28.0	44.6	65.8
Near East and North Africa	11.9	21.2	35.0	68.6	72.3	89.9
Sub-Saharan Africa[a]	9.9	9.4	13.4	28.5	29.1	33.8
Latin America and the Caribbean	31.7	53.8	76.6	80.1	110.2	139.8
East Asia	8.7	37.7	58.5	3.6	10.0	17.8
South Asia	3.9	5.3	11.7	37.0	67.5	106.9
Industrialized countries	61.5	88.2	100.1	185.5	212.2	221.0

Source: WHO https://www.who.int/nutrition/topics/3_foodconsumption/en/index4.html


Limiting factors to livestock production in developing countries
Environmental
Socio-economic
Institutional, policy & technical
Management system
High ambient temperature
Diseases and parasites
Shortage of water
Inadequate feeds (quantity/quality)
Recurrent droughts
Illiteracy
Cultural and/or religious taboos
Ethnic conflicts
Poor infrastructure
Underfunding
Undeveloped marketing system
Poorly defined / no livestock strategies
Lack of breeding policies
Shortage of skilled personnel
Weak institutions
Phenotypic & genetic parameter estimates?
Ineffective breeding programs
Traditional, subsistence
Low-input low-output
Multipurpose indigenous breeds
Few animals per herd/flock
No identification & recording system
Mobility (pastoral)
Low productivity
Low production
5

Past genetic improvement efforts

- Genetic improvement efforts in developing countries depended on breed replacement and/or crossing
- **Most interventions failed due to various reasons**
  - Incompatibility of introduced genotypes with:
  - Smallholders' breeding objectives
  - Multiple functions needed from an animal - multipurpose breed
  - Mostly subsistence oriented – not driven by market forces
- Management practices
  - Low input-low output system
  - Small herd/flock sizes
  - Environmental conditions
  - High ambient temperature
  - Fluctuating feed (amount, quality) and water supply
  - Prevalence of parasites and disease causing pathogens

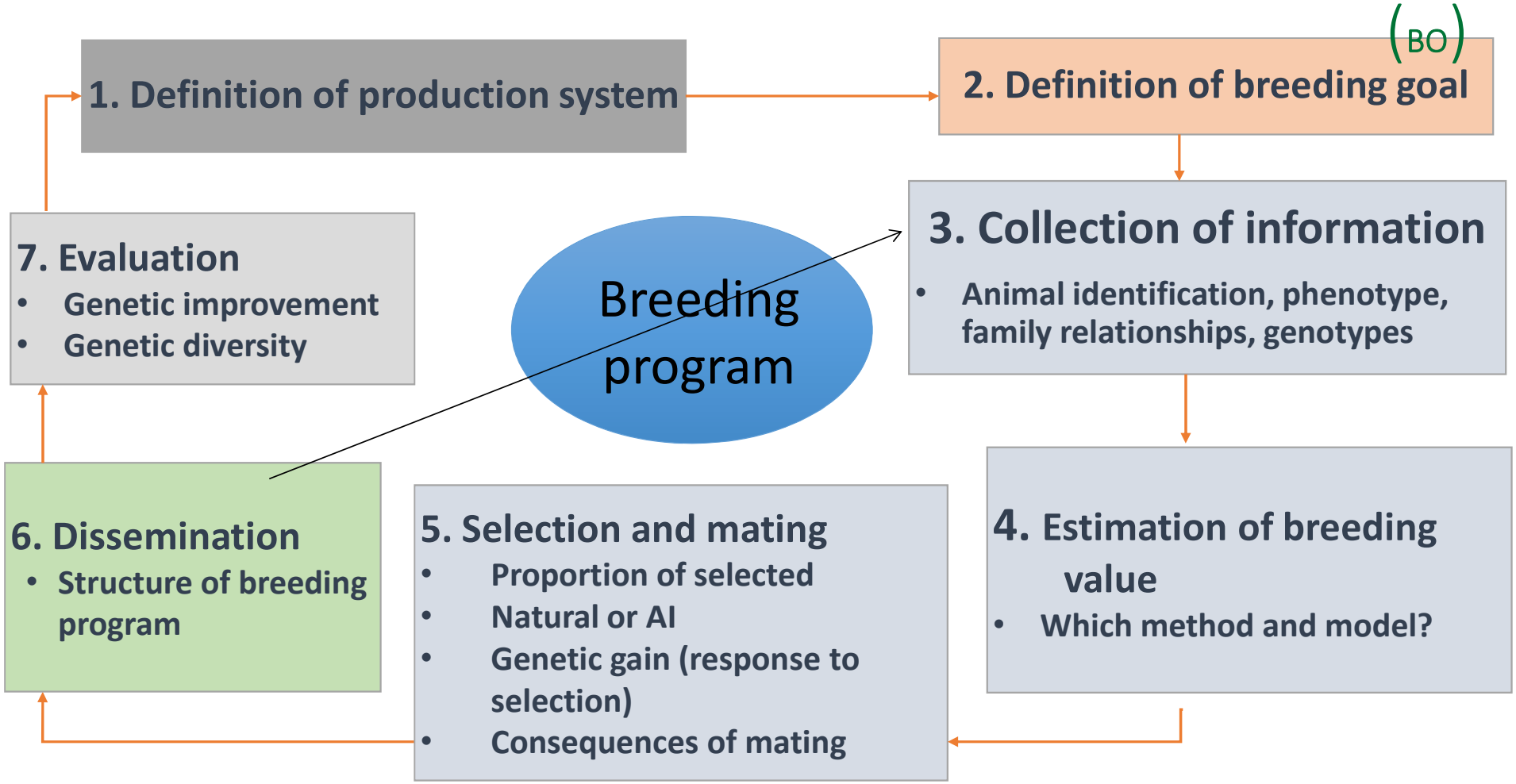
Reasons for failure of past interventions…

- Complex breeding programs that require a lot of logistics and technologies
  - Lack of comprehensive approach to design simple but effective breeding strategies
  - Blanket recommendations – were not area specific
- Indiscriminate crossbreeding of native with exotic breeds
  - Failure to maintain suitable exotic blood levels
  - No selection on the dam line
- Dependence on externally funded projects
  - Most collapsed few years post project periods

Reasons for failure of past interventions…

- Underestimating or intentionally ignoring farmers/producers
- Insufficient/no systematic breed evaluation studies
  - Fair comparison of the relative merit of indigenous and exotic breeds under typical environment was missing
  - Indigenous breeds can sometimes be superior in overall productivity and profitability
- Ignorance of the role of G x E interaction

Basic components of successful breeding programs



Basic components…

- **Breeding goal/objectives** – where should we go?
  - Which traits must be improved?
  - Economic traits? Appearance/fancy/cultural traits?
  - How important is each trait?
    - Determine economic values/weights
- Identifying animals with "best" genetics, relative to breeding goal
  - Genetic evaluation (performance recording and testing programs)
    - Which traits; on which animals?
    - Performance test station/nucleus herds?
    - Field recording?
    - Progeny testing?
- Pedigree registration

Identification of breeding goals/objectives (BO)

- Formulation of acceptable and viable breeding programs requires identification of BO in a participatory and comprehensive approach
- BOs include all relevant attributes of an animal such as production, reproduction, fitness & health characteristics
- **Importance** of each attribute depends on production circumstances
- e.g. milk Vs. meat production; fitness Vs. production traits, …
- Careful analysis of information on all aspects of the production system should give a set of BOs
- Participatory tools that aid in definition of BOs are available

Tools for definition of BOs…

- Literature review – search for published information and/or use one or combinations of these approaches
- Production system studies – system wide characterization
- Participatory rural appraisal (PRA)
- Choice model experiments
- Live animal ranking experiments
  - Own-flock ranking
  - Group-animal ranking
    - Stakeholder consultation workshops
    - Pay attention to engage all relevant stakeholders and pertinent sources of information (e.g. women and children)

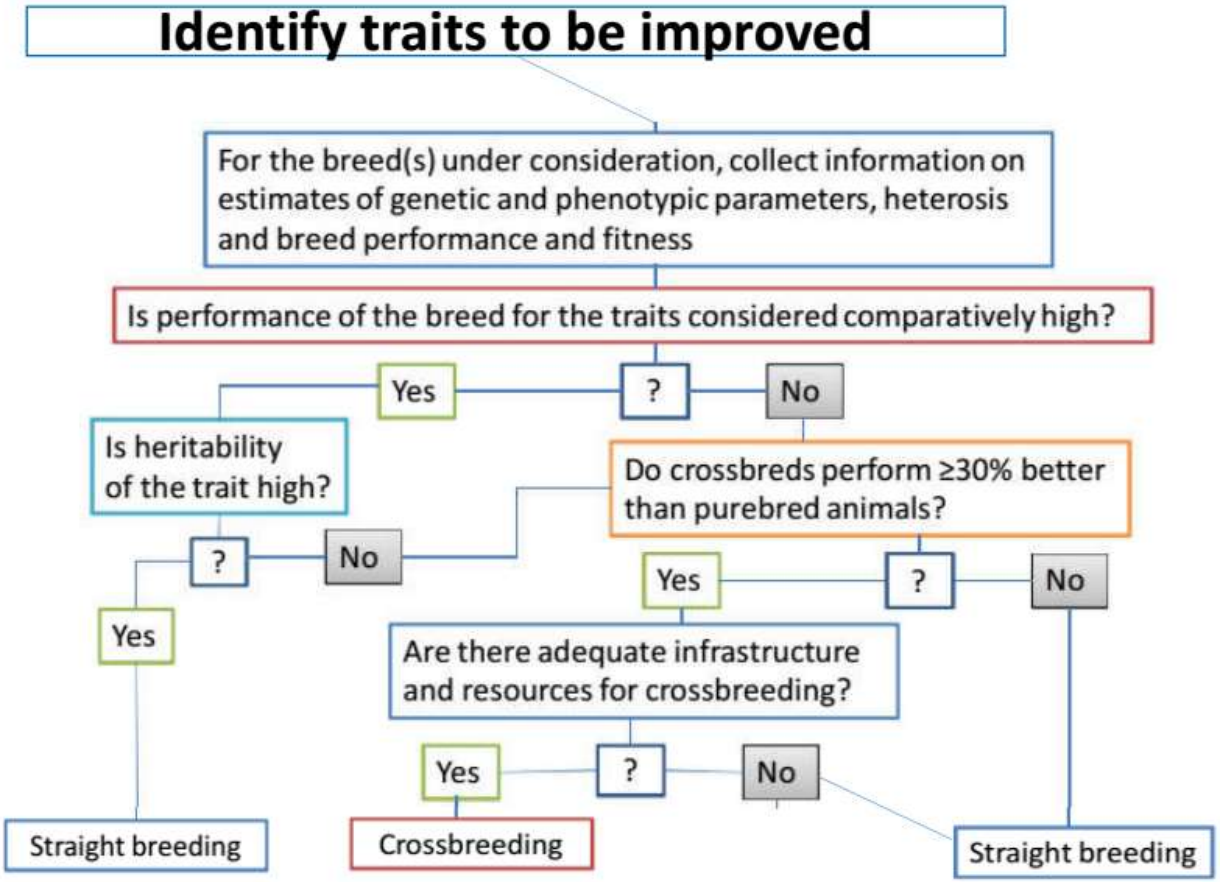
Basic components…

- **Selection and mating**
  - Mate "best" animals to produce next generation –genetic improvement
  - Determine how many and which ones to select
  - Devise robust mating system
  - Is there an option to increase number of progeny/parent by reproductive technologies?
- Sharing of breeding animals – How? To what extent?
  - Sell, dowry, gift, compensation (settling of disputes), herd splitting, borrowing, …
  - Balance rate of genetic gain and rate of inbreeding to avoid consequences of inbreeding depression

Which genetic improvement strategy?

- **Selection among indigenous population?**
  - Takes long time to realize meaningful results
  - Does not help to respond to current pressing needs
- **Replacement with exotic breeds?**
  - Adaptation related problems
  - Very expensive
- **Crossbreeding?**
  - Loss of heterosis in subsequent generations
  - Problem to maintain appropriate blood levels
- **Composite breed formation?**
  - Technically and financially demanding
  - Requires long term commitment
  - Outdated

Decision making tree to choose sound strategy



Designing breeding programs and Implementation

❖ ***Focusing on Ethiopian small ruminant breeding programs Experience***

- **Animal breeding** involves the selective breeding of domestic animals with the intention to improve desirable (and heritable) qualities in the next generation

- Selection among individuals



Mating



Towards the direction breeding goal traits



Designing breeding programs

Breeding program is the organized structure that is set up to genetically improve livestock population

Two general question must be answered in designing and implementing a breeding program

1. Where to go?
2. How to get there?

Designing breeding programs

- Success of a breeding program depend on how we perform the components of breeding program

Technical	Operational considerations
• Animal identification and recording • Performance testing • Estimation of breeding value • Selection and mating • Measuring selection response • Dissemination scheme	• Project management • When is what to do (chronological coordination of the program) • Who is doing what? Which person or organization responsible for which task in the designing and implementation phase

Remarkable results have been achieved by the developed world in well-designed organized breeding schemes





- specialized breeding farms and breeding companies which produce improved breeding stock readily available for use in commercial farms

- Strong national breeding program





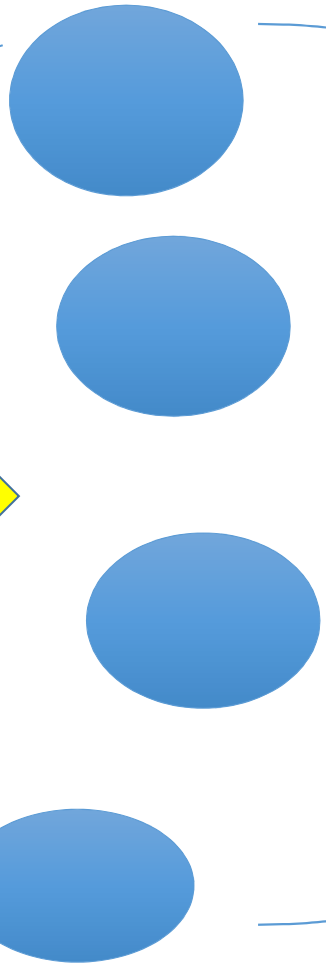


Small ruminant breeding programs in Ethiopia - Approach

Started in 1940's
Research and most ranches established since 1970's

- Breed import, test and multiplication
- Nucleus for local breeds

Disseminate improved breed/sire



Smallholder breeders flock

Nucleus

❖ *In general not successful at all stages*

Small ruminant breeding programs in Ethiopia - Approach

1. Exotic breed farms

- Dorper
  - Jigjiga
  - Werer
  - **DBARC**
  - **Areka**
- Boer goat
  - **Jinka**
  - Hawassa University
  - Haromaya
  - Adami Tulu
  - Ataye
  - Sirinka

2. Local breed farms

- **Menz** sheep improvement at DBARC
- Horro - sheep
- Afar sheep improvement at Werer Research Center
- Washera breed improvement at Adet
- Adami Tulu- Goat
- Begayit sheep and goat multiplication
- Gumuz sheep research
- Bonga sheep

Small ruminant breeding programs in Ethiopia - Approach

Challenges:

In government nucleus farm

- Technical and financial limitation
- Lacks supportive infrastructure and capacity
- Disease associated with confinement and new breed

In production unit – Smallholders

- Breed choice – preference mismatch
- Small flock size – less suitable for conventional breeding
- Low input and subsistence system
- Poor adaptability of exotic and crossbreds

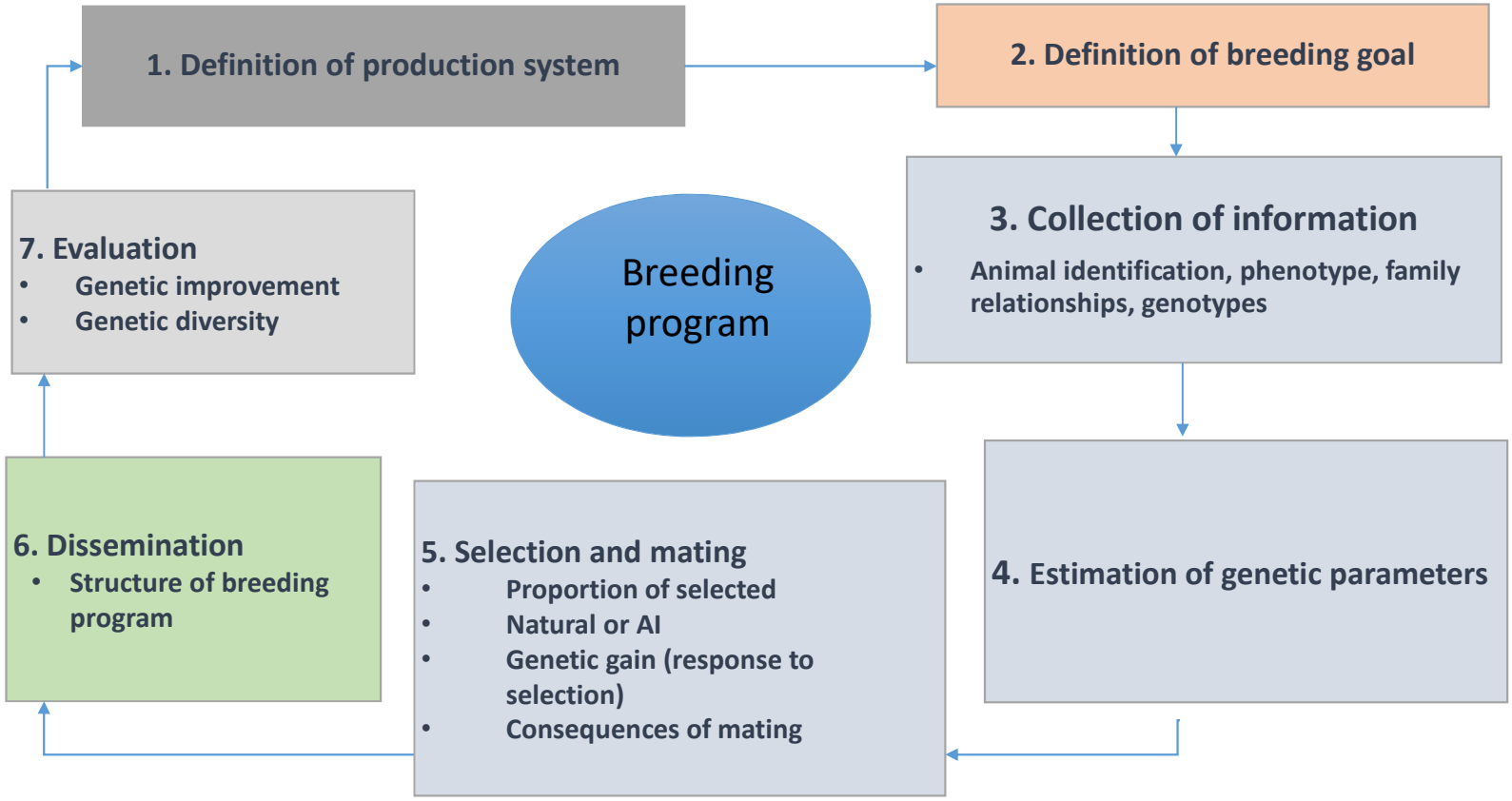
Community based breeding program (CBBP) – new thinking

- New thinking invented CBBP – started in 2008 in Ethiopia
- It mainly bases:
  - Participation
    - **Communities take a leading role** and full participation in designing and implementing the components of the breeding program
    - Allow to use and upgrading indigenous knowledge
    - Ensures community ownership
  - Organizing farmers
    - Allows to create reasonably large flock
    - Capacitate community
  - Genetic improvement and measurement directly start at community flock
    - Disease risk reduced
    - Shorten the processs
  - Based on their particular situation and capacity
    - Fits low input

Steps for setting up community-based breeding



CBBP Implementation procedure



1. Definition of production system

Breed, site and community selection

breed

- Relative contribution of the breed/population for small holder farmer/pastoralist
- Size and area coverage of the breed

Site and community

- Accessibility
- Willingness to participate
- Sufficiently large (combined) flock
  - 500 to 1000 breeding dam/per community

1. Definition of production system

Site and community (cont)

- Input and market access: Agro-industrial parks
- Government support
- Support from NGOs

Breed characterization

- Phenotypic and molecular characterization
- Genetic diversity of the breed/population

2. Definition of breeding goal –where to go?

Which trait/character of a breed need to be improved?

Decided with full participation of the community

Current and future market demand should be considered

- Thus information on current and predicted consumer demands
- Valuable market related information

2. Definition of breeding goal –where to go?

Different methods can be used



Individual interview



flock ranking

Finally results presented to the community to agree on common breeding objectives

2. Definition of breeding goal –where to go?

Trait preferences

	Menz	Horro	Bonga	Afar
Traits	• Growth performance • Lamb survival • Fleece weight	• Growth • Twining rate • Lamb survival	• Growth • TR • LS	• Milk pdn • Growth • LS

3. Collection of information

Animal identification and performance recording

- Why important from breed improvement perspective?
- In breeding program we are interested in genetic component
  - **Phenotype = Genotype + environment**
- If we manage genetically uniform animals in different environment/management the difference among individuals is completely due to environment
    - Produces a wide range of phenotypes
- When we manage different genotype in absolutely uniform environment, all difference accounted to genetic variation
  - Phenotype is be best indicator of genotype
- However, in real situation genotype and environment can not be uniform
- Thus performance records used to adjust for known environment variation and linking genetic relationship with performances enable to estimate the genetic worth or breeding value of animals for selected traits

3. Collection of information

Animal identification method





- Ear tags are the most commonly used identification methods because they are relatively cheap, easy to apply, and are less stressful to animals
- All animals of selected individuals need to be identified
- Then after offspring born should be identified at birth
- As a general rule unique number should be given for each animal and the ID given to the offspring/should be greater than the ID given for sire and dam



3. Collection of information

Performance recording



- Data to be collected should be decided based on the identified selection criteria
- Development and use of a simple, flexible, and cost-effective performance recording and evaluation system is essential
- Different format developed
  - **Dam format** – deals on lambing/kidding details (Id, sex, birth type), doe post partum weight, parity, milk yield …
  - **Lamb/kid format** - Birth weight, weaning weight, six months weight
  - **Health format**
  - **Disposal record format**
- Birth weight should be measured within 24 hours after birth
- Suspended balance can easily be used to weigh animals

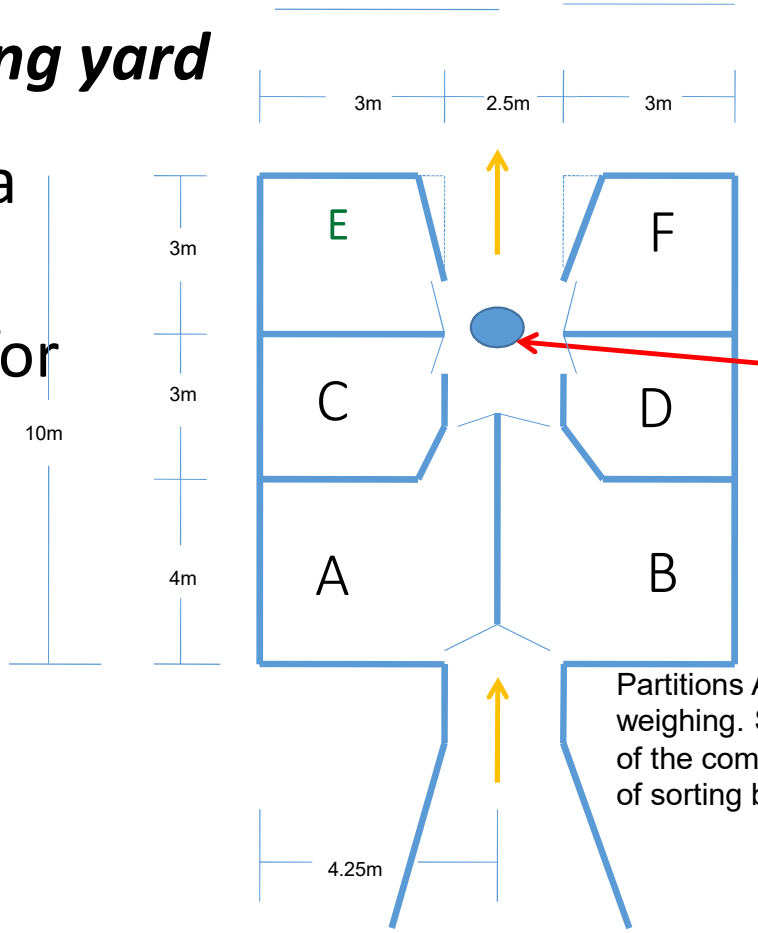




3. Collection of information

Multipurpose animal collecting yard

- Important to facilitate data collection and recording
- Helps to sort out animals for selection
- Used de-worming and vaccination



3. Collection of information

- The yard could be constructed using locally available materials with full participation of the community
- Site for collecting yard construction has to be selected by all members of the community





3. Collection of information

Recruitment of enumerators and data coder

- Selected by the community
- Enumerators need to have good conduct approved by the community, honest and committed to serve the community
- Enumerators take responsibility of animal identification, data collection and technical follow-up of the breeding program
- They served as bridge connecting the researchers and the community
- Routine (daily) monitoring and follow up flock required
- They are permanent employee in south



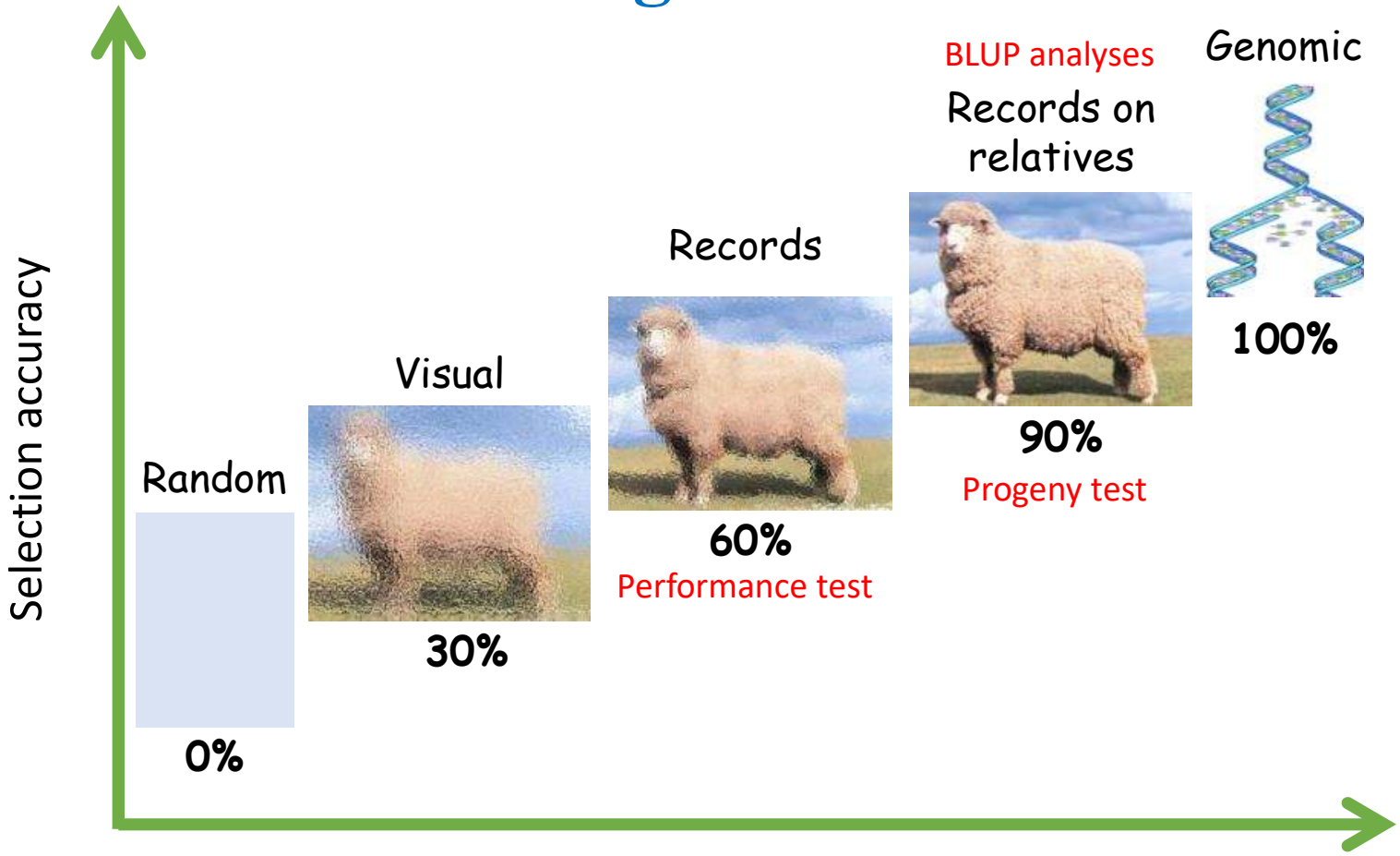


4. Estimation of breeding value
5. Selection and mating

Ranking animals

- The challenge to find the best animals as parents for the next generation
- We have observations on the phenotypes of the animals, but the phenotype is not only determined by the genetic potential, but also by environmental influences
- In order to rank the animals according to their genetic merit we need to find a way to quantify
  - We only estimate genetic potential = *Estimated Breeding Value (EBV)*
- The EBV is expressed relative to the average animal in a population

5. Selection and mating



5. Selection and mating

Selection

- Best animals need to be selected and maintained within the community
- Designing appropriate time of selection and age of the animal at selection depending on the breed and market situation is crucial
- Selecting buck lambs immediately before major festivals (New Year, Christmas, Ester, Ed-Al-Adha) are recommended

5. Selection and mating

- Selection can be undertaken at different stages
- For example, the first stage could involve culling of animals with undesirable phenotypic characteristics (e.g. tail type, coat color, horns, conformation, and general appearance) and clearly observable and genetic defects (e.g. testicle deformation)

5. Selection and mating

Community involvement in sire selection



- The community has to be actively involved in the selection process so that the ram ranking closely match their (own valuations) goals and desires
- Candidate sires should be ranked based on estimated breeding value and then farmers need to make final selection considering their own phenotypic assessment
- This helps to build trust, confidence and a sense of ownership among the beneficiary community

5. Selection and mating

Animal Show

- Animal show is an event where animals are exhibited and judged on certain phenotypical breed traits
- Animal exhibitions or shows could be linked with the ram selection events
- Importance of selection appreciated by the community
- Helps to establish market linkage
- Best animals and best breeders could be awarded to create healthy competition among members of the community
  - Best breeders would also be selected considering their animal management skill and collaboration in data collection
  - Judging should be done through a participatory process



5. Selection and mating

Sire acquiring and use arrangement

- Selected sires having best genetic merit should fetch above average income for the owner
- At the beginning the program need to have revolving fund from the project to buy best sires
- Then, the sires have to be transferred to the community
- After giving service, sires will be castrated, fattened and sold at good price
- Money obtained from the sale of culled breeding sire used to buy replacement sires

5. Selection and mating

Mating

- Use of as few as possible sires without affecting readily accessibility to all breeding dams is important in order to attain maximum genetic gain
- Different mating/breeding groups created and a selected sire allocated to each group
- Allocating farmers in to different breeding group based on their neighbourhood, communal grazing land and willingness
- If breeding sires are needed for more than one generation it is important to rotate sires among mating groups within the community to reduce inbreeding effect
- Use of AI speed up genetic gain

5. Selection and mating

Culling unselected sires

- Unselected sires should be avoided from mating
- This might be done using different methods like isolation, tethering, selling, castrating or protect mating using apron
- Imposing technologies like conditioning or fattening of unselected sired add values
- Linking them to market

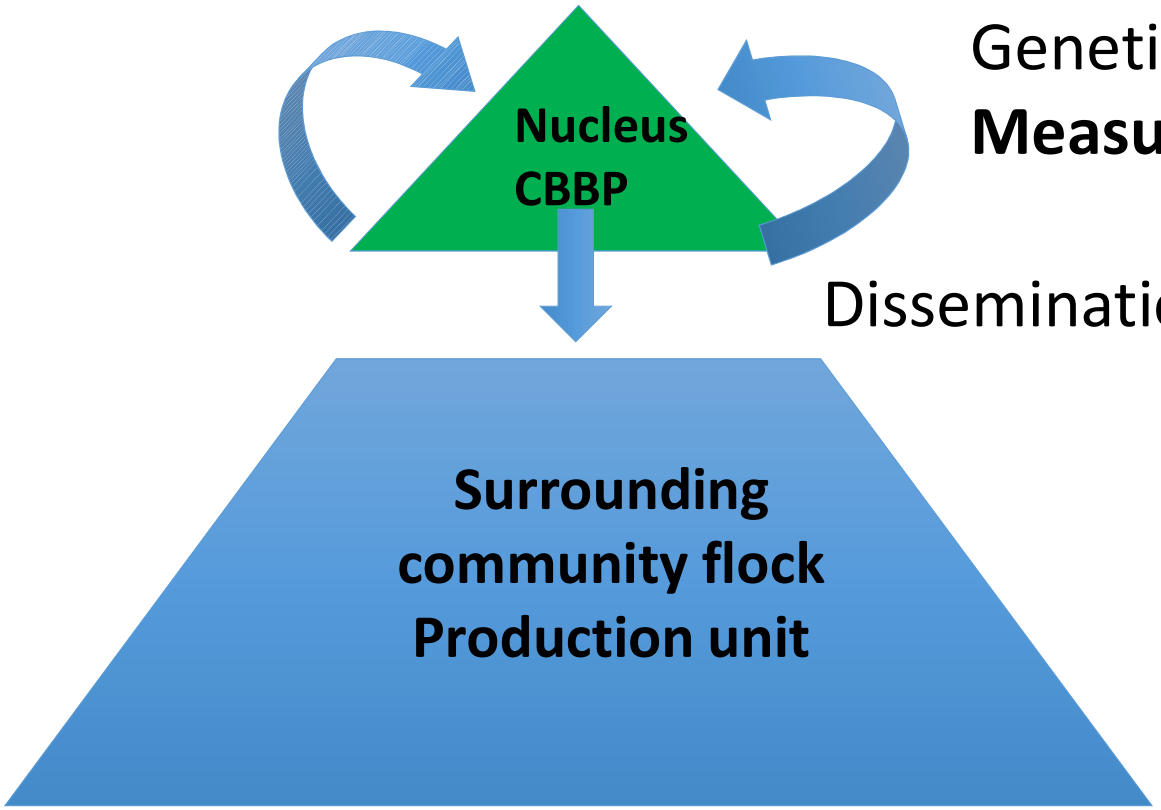
6. Dissemination

Structure of breeding program

- Genetic gain obtained in the pilot CBBP have to be disseminated to other flock
- How to sell/give improved sire to surrounding flock is challenging
- Work in communities around the CBBP to create sire market linkage
- There are promising results in using AI in the CBBP nucleus, thus extrapolating this to surrounding community based using semen from top sire need to be considered
  - Best sires served the community for one or two years became aggressive in some sites. Such sires could be maintained in central places and used for AI

6. Dissemination

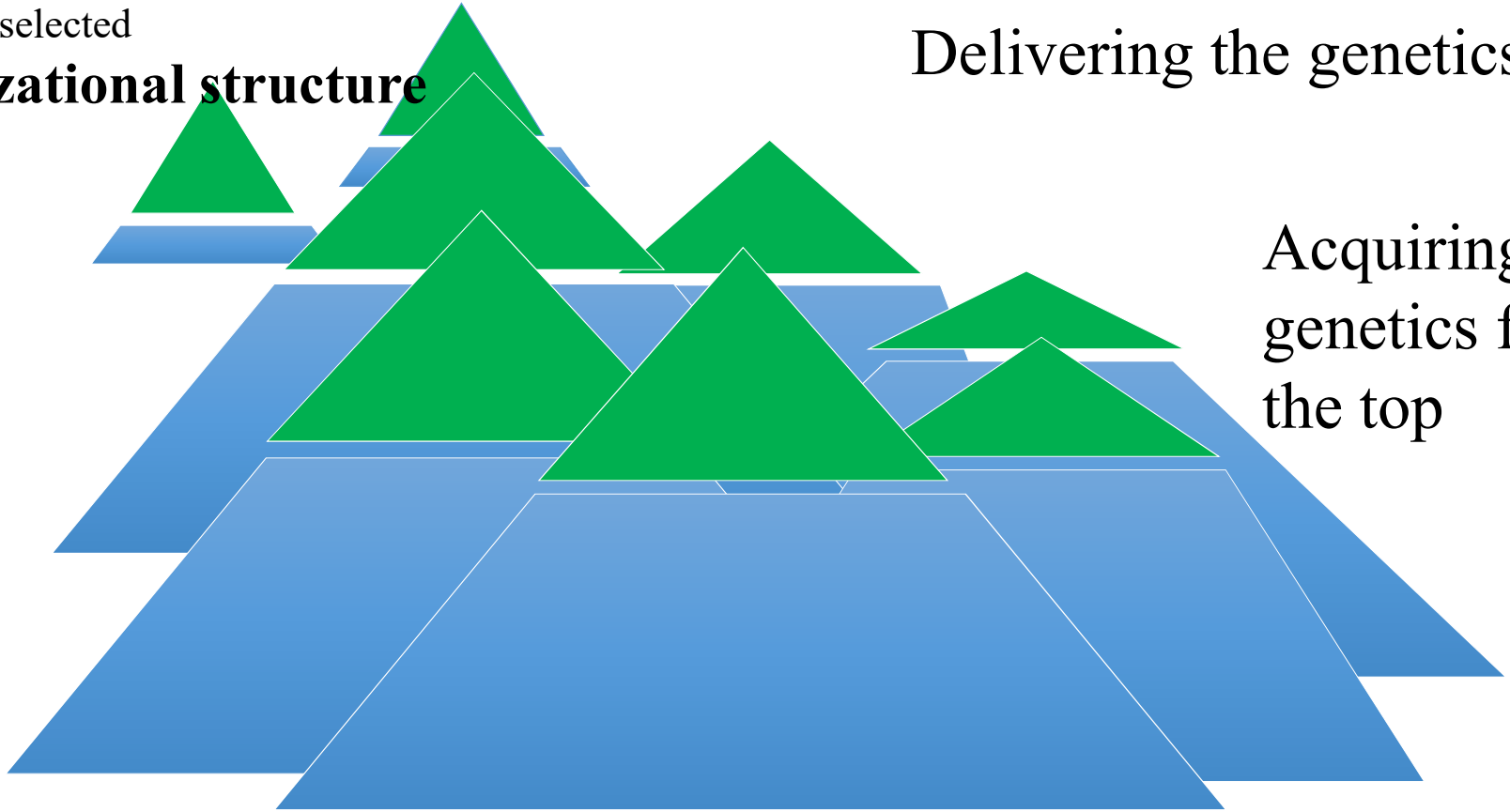
Example of two-tier breeding program



6. Dissemination

The following need to be set for each breed based on their context

- number and location of nucleus
- Number of animals in the nucleus
- Proportion selected
- **Organizational structure**



Delivering the genetics

Acquiring genetics from the top

7. Evaluation

- Success can be judged by looking shift in population average from one generation to the next
- How big is the genetic response to selection? If all went according to plan, this will resemble your predicted genetic response. However, in some situations the predicted and realized genetic response will be different
- In that case it is essential to find out and to make adjustments where needed
- Check balance between progress and genetic diversity
  - A higher selection intensity means selecting relatively few animals for breeding
    - Higher rate of inbreeding, and thus loss of genetic diversity
- Good monitoring and evaluation system should also be in place for operational implementation of breeding program